

Moamen Moustafa Basyoni

R&D Engineer — Backend, Distributed Systems, AI & Platform Engineering

Alexandria, Egypt · +20 102 786 8067 · mmoamen270@gmail.com · github.com/moamen270

PROFESSIONAL SUMMARY

Software engineer with roughly three years of production experience spanning enterprise .NET backend development, distributed-systems architecture, and applied AI/agent engineering, plus a Python backend practice built alongside the .NET work. Currently an R&D Engineer at a healthcare software company with a two-part mandate: build AI-integrated products and AI tooling across the full software delivery lifecycle, and work with the platform architect to modernize and maintain a large legacy estate. That combination produced an unusual breadth for the experience level — replacing a distributed-transaction design that had forced an entire platform onto a shared database, building an internal AI-driven SDLC automation platform surfaced through a custom Azure DevOps extension, and directing a frontend migration off a legacy Vue/AngularJS stack — alongside earlier work building a PropTech marketplace from scratch as a technical equity partner, and e-commerce/booking platforms as a first job out of university. Comfortable working in both C#/.NET and Python, across Azure and AWS, and in a regulated domain (healthcare). This document is the full profile; a CV tailored to a specific role selects and reorders from it rather than using all of it at once.

AREAS OF EXPERTISE

- Enterprise .NET backend development (C#, ASP.NET Core, Entity Framework Core)
- Python backend development (FastAPI, Flask, SQLAlchemy, Pydantic v2, async)
- Distributed systems & microservices architecture
- Legacy system modernization (.NET Framework → modern .NET; monolith → multi-repo; MS DTC → Saga)
- Agentic AI & LLM engineering (RAG, GraphRAG, multi-agent systems, MCP)
- AI-driven software delivery lifecycle (SDLC) automation
- Identity & security engineering (OIDC/OAuth2)
- API gateway, service discovery & event-driven messaging
- Payments & third-party integrations (Stripe, Paymob, webhooks/HMAC)
- Cross-cloud infrastructure (Azure and AWS)
- Healthcare / regulated-domain systems (HMIS)
- Technical enablement, mentoring & instruction

TECHNICAL SKILLS

Languages: C#, Python, SQL, JavaScript, TypeScript

.NET & Backend: .NET 8/9/10, .NET Framework 4.5, ASP.NET Core (MVC & Minimal APIs), Entity Framework Core, .NET Aspire, REST APIs, gRPC, Webhooks, XML processing

Python Backend: FastAPI, Flask, SQLAlchemy (Core + ORM 2.0), Pydantic v2, Alembic, async Python (asyncio, httpx, aiohttp), OpenAPI, pytest / pytest-asyncio, Poetry / uv / pip

Distributed Systems & Messaging: Temporal (Saga orchestration), RabbitMQ, MassTransit, gRPC, event-driven architecture, MS DTC → Saga migration

Architecture: Clean Architecture, Microservices, Service-Oriented Architecture, Layered Architecture, Business Rule Engines, API Gateway architecture, Reverse Proxy architecture

API Gateway & Networking: Nginx, YARP Reverse Proxy, load balancing, dynamic routing, health-aware routing, SSL termination, response caching

Identity & Security: OpenID Connect, OAuth2, OpenIddict, JWT (access/refresh tokens), scope-based and claim-based authorization, OWASP-aware coding, PII masking

Service Discovery & Configuration: Consul (service discovery, health checks, Consul Template, Consul KV), dynamic service registration/deregistration, centralized configuration

Databases: SQL Server, PostgreSQL (relational) · Neo4j (graph)

Search & Retrieval: GraphRAG, contextual retrieval, query expansion, semantic search, Elasticsearch

AI & Agentic Systems: LangGraph, CrewAI, LangChain, Model Context Protocol (MCP) — five custom MCP servers built (Azure DevOps, Figma, XMind, Playwright, database); RAG and GraphRAG; multimodal search and retrieval (image, location, and natural-language query); multi-agent and cyclic agent workflows; tool/function calling

LLMOps & Observability: LangSmith, LangFuse, OpenRouter organization management, OpenAI organization management, token usage tracking, cost optimization, model selection / multi-model orchestration

AI Security & Guardrails: prompt-injection mitigation, LLM policy enforcement, human-in-the-loop review, PII masking, rate limiting, token budgeting, prompt telemetry

Web / Frontend: JavaScript, TypeScript, jQuery, Ajax, HTML, CSS, XML. **Angular** — working knowledge; directed a migration to it as the target frontend architecture, not a hands-on production Angular developer. **Vue 3 / AngularJS / PrimeNG** — familiarity from the same migration (legacy and target stacks), not development experience. **React** — working proficiency, self-assessed, no documented production project. Micro-frontend / shell architecture — architectural direction, not implementation.

Payments & Integrations: Stripe, Paymob, webhook processing, HMAC signature verification, Google Sheets API (two-way), Google Drive API, Microsoft Graph / SharePoint, SMS and email notification providers

DevOps & Version Control: Azure DevOps (cloud & on-premise, Pipelines, Artifacts, custom extension development), GitHub Actions, CI/CD automation, TFS/TFVC → Git migration, monolithic repo → multi-repository decomposition, Git Flow, Docker, private NuGet feed, private npm feed

Cloud & Hosting: Azure — App Service (Web Apps), Blob Storage, Azure SQL, Static Web Apps, DevOps Pipelines/Artifacts. **AWS** — EC2, S3,

messaging services (SMS/email notifications). Cross-cloud evaluation and cost optimization; IIS hosting.

AI-Assisted Development Tools: Claude Code (daily), GitHub Copilot, Codex, OpenCode

Performance: BenchmarkDotNet, performance tuning, legacy system modernization

Training & Instruction: .NET, Generative AI, AI Agents — delivered through Andalusia Academy

PROFESSIONAL EXPERIENCE

R&D Engineer — Backend, Distributed Systems & AI Engineering

Andalusia Health Business Solution (AHBS) · Alexandria, Egypt · December 2023 – Present *Healthcare / HMIS, serving the Andalusia hospital group in Egypt and Saudi Arabia* · 5-developer delivery team · Agile Scrum on Azure DevOps

The role carries two mandates: building AI-integrated products and AI tooling across the software delivery lifecycle, and working with the platform architect to analyze, modernize, and maintain the core platform — then implementing, reviewing, deploying, and supporting that plan alongside the development team as a senior developer.

AI & R&D Engineering - Built five custom Model Context Protocol (MCP) servers (Azure DevOps, Figma, XMind, Playwright, database), surfaced to the organization through a custom Azure DevOps extension so the whole AI toolchain runs inside one platform. - Built an AI-driven Software Delivery Lifecycle (SDLC) automation program: AI-assisted requirements analysis and user-story authoring grounded in a business GraphRAG knowledge base, automated story-quality scoring, Use Case Point (UCP) story weighting, AI-generated bug reports and test cases, and an automated pull-request reviewer that opens threads, posts overall and line-level comments, and checks implementations against the business requirements in the linked work item. - Built a code graph spanning 38 modules of the production application for code-to-business traceability, an AI code-documentation system combining Roslyn static analysis and code metrics with LLM interpretation, and a documentation portal with an embedded AI assistant. - Delivered internal AI products for specific business functions: a mail AI assistant for C-level and managerial users, an assessment chat that generates and scores multiple-choice/essay quizzes for HR and Learning & Development, and a conversational clinic-booking assistant for the Customer Experience team. - Built RAG and GraphRAG systems on Neo4j, including a policies knowledge base and a PII-masked organizational memory graph; built LangGraph workflows and CrewAI multi-agent pipelines; implemented multi-model orchestration across OpenAI and OpenRouter with observability via LangSmith and LangFuse. - Managed OpenRouter organization infrastructure — model access, provider configuration, usage/cost monitoring, and rate limiting/token budgeting. - Applied AI guardrails: prompt-injection mitigation, human-in-the-loop review, and PII masking.

System Modernization Program (with the platform architect) - Analyzed a platform running .NET Framework 4.5 and ASP.NET Core 2.2 side by side, with distributed transactions handled through a custom MS DTC workaround that forced the entire system onto a single shared database, and source control on one monolithic TFVC repository on a legacy TFS server. - Replaced the MS DTC workaround with **Temporal Saga orchestration**, improving 10+ inter-service communication patterns. - Migrated source control from legacy TFS/TFVC to Azure DevOps Git, decomposed the monolith into multiple Git repositories, and stood up a private NuGet feed for shared packages. - Modernized 5 core and 7+ additional modules off the legacy stack, applying Clean Architecture. - Designed the target architecture — gRPC for inter-service communication, MassTransit messaging, service discovery, centralized configuration with secret management, an OIDC/OAuth2 identity server, and an API gateway — and implemented an identity server (OpenIddict, JWT, scope/claim-based authorization), Consul-based service discovery/dynamic routing/centralized configuration, and an Nginx API gateway with parallel service versioning that cut deployment downtime from **2 hours to 15 minutes (87.5%)**. - Containerization (Docker and Kubernetes) is planned and deliberately deferred until the work above is complete and stable — **not yet implemented**.

Frontend Modernization — technical direction (not hands-on development) - Defined and directed the migration from a monolithic Vue 3 shell embedding multiple AngularJS applications through iframes, to a single modern Angular shell application on PrimeNG, split into multiple repositories distributed through a private Azure DevOps npm feed. - This was a planning and oversight role — setting the target architecture and directing delivery; hands-on Angular development was not part of this work.

Backend & Platform Engineering - Built and maintained server-rendered and API-driven web applications using ASP.NET Core MVC and Minimal APIs, with JavaScript/jQuery/Ajax front-end behavior and XML-based data exchange, hosted on IIS. - Developed internal Python services: FastAPI (Pydantic v2, dependency injection, async/await, OAuth2 JWT security) for LLM service orchestration; Flask (blueprints, extensions) for legacy-system integration and data processing; SQLAlchemy Core and ORM 2.0 (async sessions, Alembic migrations); asyncio/httpx for external API communication; pytest and pytest-asyncio for test coverage; batch data-transformation pipelines. - Built a runtime business-rule engine, improving execution performance by **45%** (measured with BenchmarkDotNet). - Implemented AABB-aligned international blood-transfusion workflows for the Blood Bank module, reaching a **98%** compliance score, alongside Supply Chain, Bed Management, and Hospital Structure modules. - Established Azure DevOps CI/CD pipelines across Dev, Staging, and Pre-Live for both backend and frontend projects, on multi-repository Git Flow. - Mentored and onboarded **25 engineers across 6 teams over 8 sessions** on architecture and AI-assisted development.

Technical Partner (equity) — Trastain

Cairo, Egypt (Remote) · Part-time · February 2025 – August 2025

Joined a friend's early-stage company as technical partner, compensated in equity rather than salary, on a PropTech / short-term-rental marketplace.

- Analyzed existing short-term-rental platforms (Airbnb and similar), identified that the Airbnb model is not culturally aligned with the Egyptian, wider Middle Eastern, and Muslim market, and wrote the product case for a differently-shaped alternative — the platform's culture-fit matching and policy features exist because of this analysis.
 - Built the resulting vacation/short-term-rental booking platform from scratch: shared-property booking, culture-fit/preference matching, and policy enforcement.
 - Developed RAG-based AI property matching with actionable booking recommendations, and **multimodal AI search** — users can search by image, by location, or by natural-language description.
 - Handled transaction security for booking execution; integrated Stripe and Paymob payment gateways with webhook processing and HMAC signature verification.
 - Built GitHub Actions CI/CD pipelines deploying backend services to Azure Web Apps and the frontend to Azure Static Web Apps.
-

Software Engineer — MYM (Make Your Miracle)

Giza, Egypt · June 2023 – October 2023

First role after graduating, at a client-project software house delivering SaaS / e-commerce / booking systems.

- Developed backend systems for e-commerce and booking platforms across several client domains, including hotel booking, gym/fitness booking, and other reservation systems.
- Worked end to end on client delivery: requirements analysis, implementation, deployment, and support.

Freelance / Independent Client Work

With a small partner group · Side work alongside employment

An attempt to formalize freelance client work into a company (branding, web presence, an intake process) — the company was not fully established, but client delivery was real, for small offices and training centres.

- Built **Vortex**, a management system for a children’s programming and robotics training centre: course materials and programme structure, instructor session logging with management approval/denial, a management report calculating instructor pay from approved sessions, two-way Google Sheets integration against the centre’s existing sheets, and a Google Drive-powered public gallery of the centre’s work on the landing page.
- Evaluated **Azure vs. AWS** per project and chose based on delivery speed and cost: Azure managed services (Blob Storage, App Service, managed SQL, Static Web Apps) for fast MVP delivery with no self-managed servers; AWS EC2/S3 self-managed infrastructure for later projects. Replaced a cloud SMS notification service with a local Egyptian provider, cutting messaging cost.
- Integrated Stripe and Paymob payment gateways with webhook processing and HMAC signature verification.
- Built GitHub Actions CI/CD workflows deploying to Azure Web Apps and Azure Static Web Apps.

NOTABLE PROJECTS & INITIATIVES

Project	Description	Result / Evidence
AI-Driven SDLC Automation Program	AI + MCP tooling across requirements, design, QA and development, surfaced through a custom Azure DevOps extension	End-to-end delivery-lifecycle automation
Application Code Graph	Code graph spanning the full application	38 modules mapped
AI Pull-Request Review	Opens threads, overall + line-level comments, checks implementation against linked business requirements	Automated code and requirements review
Custom MCP Servers	Azure DevOps, Figma, XMind, Playwright, and database MCP servers	Reusable AI tooling across the SDLC
Mail AI Assistant / Assessment Chat / Clinic Booking Chat	Internal AI products for C-level, HR/L&D, and CX teams respectively	Delivered to production users
Distributed Transactions	Temporal Saga orchestration replacing a custom MS DTC workaround	10+ service communication patterns improved
System Modernization	Legacy .NET Framework 4.5 / ASP.NET Core 2.2 → modern .NET, Clean Architecture	5 core + 7+ additional modules
Source Control Migration	Legacy TFS/TFVC monolith → Azure DevOps Git multi-repo + private NuGet feed	Project coupling broken across repositories
Frontend Modernization (direction)	Vue 3 shell with iframed AngularJS → modern Angular shell on PrimeNG, split into repos	Monolithic frontend decomposed
Deployment Infrastructure	API Gateway with parallel service versioning	Downtime reduced 2h → 15m (87.5%)
Business Rule Engine	Runtime-injected business logic	45% performance improvement
Blood Bank Module	AABB-aligned international transfusion workflows	98% compliance score
Team Onboarding Program	Architecture and AI-assisted development training	25 engineers across 6 teams, 8 sessions
Trastain Booking Platform	Built from scratch as technical equity partner; RAG matching + multimodal AI search	0→1 product build, market analysis to launch
Vortex	Training-centre management system with approval workflow and Google Sheets/Drive integration	Freelance client delivery

EDUCATION

B.Sc. Computer Engineering — Behera Higher Institute (BHI), Alexandria, Egypt Graduated 2023 · GPA 3.2/4.0 (B+)

PUBLICATION

IoT-Enabled E-Prescription Management and Dispensing Machine Monitoring — IEEE, 2024

LANGUAGES

Arabic (Native) · English (B2, Upper Intermediate)